ELTEI501 – Install electromechanical equipment of power plant
DATE: .../.../2025

PERIOD: .....

TECHNICAL SECONDARY SCHOOLS (TSS)

END OF TERM I EXAMINATIONS: 2025-2026

QUESTION PAPER

SECTOR: Energy

TRADE NAME: Electrical Technology

RQF LEVEL: 5

MODULE TITLE: – Install electromechanical equipment of power plant

MARKS: **...../100**

DURATION: 3 HOURS

INSTRUCTIONS TO CANDIDATES:

This Exam paper is composed of Twenty-four (24) questions distributed in Three Sections (A, B, C). Follow the instructions given below, and answer the indicated questions in each section for a total of 100marks

Section **A**: Between (11-17) questions, all are **Compulsory**. **55 marks**
Section **B**: Five (5) questions, choose and answer any three (3). **30 marks**
Section **C**: Two (2) questions, choose and answer any one (1) **15 marks**

Allowed Materials: Blue pen, Ruler, Calculator, Pencil, (Fitting with the exam paper)

Choose the correct answer.

1. Which component converts **mechanical energy into electrical energy** in any power plant?
   A. Turbine

   B. Generator
   C. Condenser
   D. Penstock

2. A **diesel power plant** operates based on which principle?
   A. Rankine cycle
   B. Internal combustion
   C. Hydraulics
   D. Nuclear fission
3. A **Kaplan turbine** is best suited for:
   A. High head, low flow
   B. Medium head
   C. Low head, high flow
   D. Very high head
4. A **P&ID drawing** focuses on:
   A. Chemical composition
   B. Piping, instrumentation, and flow direction
   C. Electrical wire gauges
   D. Mechanical tolerances
5. Which hazard involves exposure to **high-pressure steam or hot fluids**?
   A. Mechanical hazard
   B. Electrical hazard
   C. Fluid hazard
   D. Chemical hazard
6. Which hazard control method involves replacing hazardous materials with safer ones?
   A. Elimination
   B. Substitution
   C. Engineering controls
   D. PPE
7. Installing flue gas desulfurization (FGD) units is an example of:
   A. Noise control
   B. Emission control
   C. Fire prevention
   D. Flood control
8. Which tool is SPECIFICALLY used to align turbine rotors during installation?
   A. Multimeter
   B. Turbine alignment tool
   C. Hydraulic jack
   D. Grinder

9. Which site preparation activity involves leveling ground and preparing foundations?
   A. Marking of areas
   B. Grading and leveling
   C. Initial cleaning
   D. Documentation
10. Which ventilation type uses fans, blowers, and ducts to move air?
   A. Natural ventilation
   B. Hybrid ventilation
   C. Mechanical ventilation
   D. Passive ventilation

TRUE OR FALSE (20 MARKS)

Write *TRUE* or *FALSE*.

1. Hydroelectric power plants convert the potential and kinetic energy of water into electricity.
2. Diesel power plants cannot be used in remote areas without grid connection.
3. Steam power plants operate using the Rankine cycle.
4. Electromechanical drawings always show real-world dimensions.
5. Personal Protective Equipment (PPE) is the first line of defense against hazards
6. Elimination is the most effective hazard control method because it completely removes the hazard.
7. Safety signage is unnecessary if workers are highly trained.
8. Floodgates and water diversion channels are examples of flood control in power plants.
9. Cranes and hoists are used for testing electrical equipment during installation.
10. Site visits help identify risks, assess utilities, and evaluate environmental conditions.

FILL IN THE BLANKS (15 MARKS)

Write the correct word or phrase.

11. A facility designed to convert different forms of energy into electrical energy is called a _____.
12. A thermal power plant uses _____ energy to generate electricity.
13. The water passage that directs flow to the turbine in a hydro plant is called the _____.
14. The rotating part of a generator is called the _____.
15. The plant component that removes excess heat is the _____ system.

Section **B**: Five (5) questions, choose and answer any three (3). **30 marks**

1. How would you explain the main purpose and operation of a power plant to someone who has no prior technical background?

2. When reviewing electromechanical component drawings, what steps do you take to ensure correct interpretation and avoid installation errors?

3. What potential hazards do you consider most critical during pre-installation activities in a power plant, and how do you plan to prevent them?

4. What factors influence your decision when selecting tools, materials, and equipment for power plant installation work?
5. How do you assess and prepare a site to ensure it meets all technical and safety requirements before beginning power plant installation?

Section **C**: Two (2) questions, choose and answer any one (1) **15 marks**

1. Clearly explain Merits and Demerits of diesel power plant

2. Fill the name of the main components of a hydropower plant

